A Multitask Learning Attention-Based Hybrid Model with Pre Trained Embeddings for Mental Health Prediction

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***Abstract*—Mental health disorders such as depression, anxiety, and suicidal ideation have become pressing concerns in today's digital age. Early detection of such conditions using automated methods can enable timely interventions. In this study, we propose a multi-task learning (MTL) attention-based hybrid model leveraging pretrained GloVe embeddings to predict mental health conditions from text data. The model utilizes a shared feature extraction mechanism based on a Bidirectional Long Short-Term Memory (BiLSTM) network and an additive attention mechanism, which enables capturing crucial contextual information. Task-specific layers are implemented for three classification objectives: depression detection (binary), suicidal ideation detection (binary), and a five-class mental health head covering anxiety, bipolar disorder, normal, personality disorder, and stress. The model is trained on three public Kaggle datasets using a stratified train/validation/test split in which duplicate texts are removed before splitting, so that no test text is seen during training. We compare the MTL model with single-task learning (STL) RNN, CNN, LSTM, and BiLSTM-attention baselines using accuracy, macro-F1, and AUC-ROC measured on the held-out test set. On the test set, the MTL model reaches accuracies of 93.32, 78.20, and 92.80 for depression, mental health, and suicide, respectively. The findings highlight the potential of multi-task deep learning frameworks for AI-assisted mental health assessment tools.**

***Index Terms*—Anxiety classification, Attention mechanism, BiLSTM, Deep learning, Depression detection, GloVe, Mental health prediction, Multi-task learning, NLP, Pretrained embeddings, Suicidal ideation.**

I. Introduction

Mental health issues have seen a global rise, with conditions such as depression, anxiety, and suicidal ideation affecting millions of people. Conventional diagnosis relies heavily on clinical evaluations and self-reported symptoms, leading to delays in treatment. The emergence of Natural Language Processing (NLP) and Deep Learning (DL) has enabled automatic identification of mental health patterns from text-based data sources such as social media posts. Multi-task learning (MTL) has gained popularity as an effective paradigm in deep learning, allowing models to learn multiple related tasks while sharing representations. This study extends prior work on hate speech detection and emotion classification by adapting an MTL framework for mental health prediction.

Key Contributions: Designing an MTL framework using a shared BiLSTM encoder and an attention mechanism with three task-specific output heads. Incorporating pretrained GloVe embeddings for better contextual understanding. Applying a leakage-free evaluation protocol (duplicate removal, stratified train/validation/test split, tokenizer fitted on training data only). Evaluating the model on multiple datasets and comparing MTL performance against single-task learning (STL) baselines.

II. Related Work

A. Mental Health Prediction Using NLP

Numerous studies have explored NLP-based mental health prediction using traditional machine learning (ML) and deep learning models. Early approaches relied on lexicon-based sentiment analysis and handcrafted linguistic features, which often lacked generalization across datasets. Modern DL techniques, including Convolutional Neural Networks (CNNs) and Transformers, have significantly enhanced predictive performance [6].

B. Multi-Task Learning in NLP

MTL has been successfully applied in NLP for tasks such as hate speech detection, emotion classification, and medical diagnosis. The ability to share learned representations across tasks improves generalization and mitigates data scarcity issues [7].

C. Pretrained Embeddings for Text Classification

Recent advancements in pretrained embeddings, such as Word2Vec, GloVe [9], FastText, and BERT, have revolutionized text classification by capturing semantic relationships between words. These embeddings are crucial in understanding complex mental health-related language patterns [8]. In the field of mental health, researchers are increasingly using natural language processing (NLP) to interpret the vast amounts of writing generated online [3]. NLP techniques are used to understand a person's mental state by analyzing everyday language from social media, blogs, and forums [5]. Detecting symptoms of depression or suicidal thoughts [3] often relies on machine learning and deep learning approaches, which have been shown to be particularly effective for these complex health-related NLP tasks in many studies [3]. This approach creates new opportunities for tracking the mental health of the general population [5]. In this case, multi-task learning (MTL) is a promising method [1]. Rather than training separate models for discrete tasks, MTL simultaneously tackles related problems such as identifying sentiment, emotion, and time references within a text [1]. Ghosh et al. [1] found that an MTL method using a specialized attention network performed better than models trained on emotions alone for emotion detection in suicide notes. Learning these related tasks together using shared feature representations helps capture a more holistic view of the situation [1].

Attention is a key component in helping systems identify the most relevant words or phrases for a particular task [4], [11]. Whether analyzing sentiment about a specific product feature or identifying important statements in a suicide note [4], it is useful. However, attention mechanisms can become too focused, essentially "overfitting" to some parts of the text and ignoring other important cues [4]. Bao et al. addressed this problem by exploring ways to "control" attention, encouraging the model to consider a larger context, which in turn improved results [4]. While deep learning models are powerful, they can sometimes act as "black boxes" and lack robustness; it is difficult to correct a specific error without retraining the entire system [4]. To solve this problem, some researchers integrate external knowledge, such as sentiment dictionaries (lists of words with known emotional polarity) [3]. From these studies, we see common challenges: single-task models can be limited [1], attention can be too narrowly focused [4], end-to-end systems can lack robustness [4], and informal, non-clinical text analysis presents unique difficulties [5]. Our research, "A Multitask Learning Attention-Based Hybrid Model with Pre-Trained Embeddings for Mental Health Prediction", directly addresses these issues. We aim to create more accurate and reliable models for predicting mental health status from text.

III. Methodology

A. Problem Formulation

- Given an input text sequence X, the model makes three predictions:
  - Y_1 – Depression presence (binary: depressed / not depressed).
  - Y_2 – Mental health condition (five classes: Anxiety, Bipolar, Normal, Personality disorder, Stress).
  - Y_3 – Suicidal ideation presence (binary: suicidal / non-suicidal).
- The objective is to minimize the combined multi-task loss function:

$$L_{\text{total}} = \alpha L_{\text{depression}} + \beta L_{\text{mental}} + \gamma L_{\text{suicidal}}$$

where L represents the task-specific sparse categorical cross-entropy loss (softmax output), and α, β, γ are weighting factors, set to 1 in our experiments.

B. Dataset Description

We have used three datasets for our specific task. All of them are taken from Kaggle. The first dataset is the suicide and depression dataset created by Nikhileswar Komati. This dataset is binary valued with two unique classes, suicide and non-suicide. The second dataset is the Depressive/Non-Depressive Tweet Data created by Mexwell. The dataset is already encoded as 0 and 1, where 0 means non-depressive and 1 means depressive. The third dataset is Sentiment Analysis for Mental Health created by Suchinita Sarkar. The dataset is a large dataset containing 7 classes: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar and Personality disorder. From this dataset we use the five classes Anxiety, Bipolar, Normal, Personality disorder and Stress for the mental health task, since depression and suicidal ideation are covered by the other two heads. Collecting posts directly may create privacy and security concerns, so building the tasks from these public datasets enabled us to prepare our own task-specific data. Null values and duplicate texts were removed. To limit class imbalance, the depression and suicide datasets were capped at 6,500 samples per class and the Normal class of the mental health dataset was capped at 2,718 samples. Each dataset is then split in a stratified manner into 70% training, 15% validation and 15% test sets.

C. Model Architecture

The model is designed for text classification in the mental health field, covering depression, suicide, and general mental health conditions. It is trained on multiple benchmark datasets, including Reddit-based mental health posts and Kaggle suicide and depression datasets. To ensure consistency of the dataset schema, we standardize columns. To balance the datasets and reduce the effect of class imbalance, majority classes are capped and class-balanced sample weights are used. Missing values are removed. Input text is then processed by stopword removal, tokenization and padding, which ensures equal input length for the model. A label encoder converts categorical labels into numerical form for model training. Pretrained GloVe embeddings convert words into dense vector representations capturing semantic relationships in text. Contextual and sequential dependencies are captured in both the forward and backward directions using a bidirectional LSTM (BiLSTM) [10], [13]. An additive attention mechanism [11] is used to identify and emphasize the most relevant words that contribute to mental health indicators. The same shared encoder is applied to the input of every task, and three separate task-specific dense layers perform depression, mental health, and suicidal ideation detection. Each task-specific output layer uses a softmax activation.

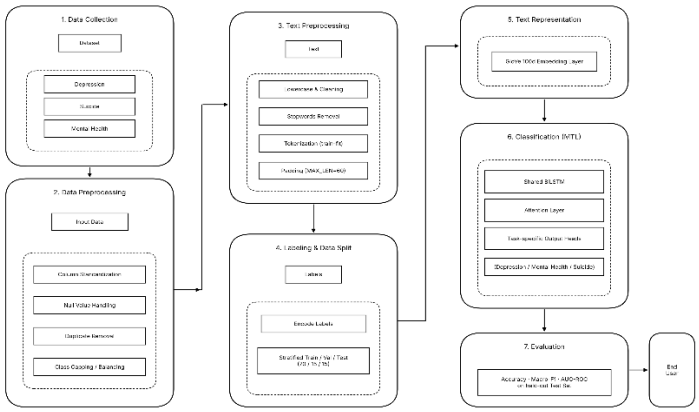


Fig. 1. Visual representation of the described model architecture.

The proposed model consists of:

- *InputLayer* – Tokenized and preprocessed text input (60 tokens).
- *EmbeddingLayer* – Pretrained 100-dimensional GloVe embeddings (50,000-word vocabulary), fine-tuned during training, with padding masked.
- *SharedFeatureExtractionLayer* – A BiLSTM (64 units per direction, dropout 0.2) processes sequential dependencies in both directions.
- *AttentionMechanism* – Additive, mask-aware attention (64 units) highlights important words relevant to mental health prediction and pools the sequence into a single vector, followed by dropout of 0.5.
- *Task-SpecificDenseLayers* – Three separate dense layers (64 ReLU units each) for depression, mental health, and suicidal ideation detection.
- *OutputLayer* – Softmax activation for each task (2, 5 and 2 classes, respectively).

D. Data Preprocessing

Text Cleaning – Lowercasing and removing URLs, special characters, digits, emojis, and stopwords such as “the”, “a”, “an”, or “in”, which makes it simpler to categorize texts into distinct groups. Tokenization & Padding – A Keras tokenizer fitted on the training split only converts texts into sequences, which are padded or truncated to a fixed length of 60 tokens. Duplicate Removal – Duplicate texts are removed before splitting to avoid leakage between training and test sets. Class Balancing – Capping of majority classes, resampling of the smaller datasets to the size of the largest one within the training and validation splits only, and class-balanced sample weights in the loss. Labelling – Labels are encoded using a label encoder, with a separate encoder per task.

E. Training Strategy

Optimization – Adam optimizer [12] with an initial learning rate of 0.001 and a ReduceLROnPlateau scheduler (factor 0.5). Loss Function – Class-weighted sparse categorical cross-entropy per task. Batch Size – 64; Epochs – up to 10 with early stopping on validation loss (patience 3), restoring the weights of the best epoch. In our run, training stopped at epoch 8 and the weights of epoch 5 were restored. Baselines – RNN, CNN, LSTM, and BiLSTM-attention models are trained as single-task models on the same splits and preprocessing.

IV. Results

The model is evaluated on the held-out test set, which is never used for training, validation, or model selection, using precision, recall, F1-score, accuracy, macro-F1 and AUC-ROC.

Key findings include: MTL achieves higher/comparable macro-F1 than the STL baselines on the depression and suicide tasks; the attention-based models outperform the plain RNN, CNN and LSTM baselines.

Classification Report for Depression:

	precision	recall	f1-score	support
Non-Depressive	95.00	91.20	93.00	975
Depressive	91.50	95.20	93.30	975
accuracy			93.20	1950
macro avg	93.20	93.20	93.20	1950
weighted avg	93.20	93.20	93.20	1950

Classification Report for Mental Health:

	precision	recall	f1-score	support
Anxiety	88.80	83.70	86.20	539
Bipolar	85.60	84.30	84.90	375
Normal	91.70	86.30	88.90	408
Personality disorder	49.60	82.80	62.00	134
Stress	72.30	66.20	69.10	343
accuracy			81.00	1799
macro avg	77.60	80.60	78.20	1799
weighted avg	82.70	81.00	81.50	1799

Classification Report for Suicidal:

	precision	recall	f1-score	support
non-suicide	94.60	90.90	92.70	975
suicide	91.20	94.80	93.00	975
accuracy			92.80	1950
macro avg	92.90	92.80	92.80	1950
weighted avg	92.90	92.80	92.80	1950

TABLE I

Comparison of MTL BiLSTM-Attention with STL Models across Tasks (Test Set)

Category	Metric	MTL BiLSTM-Att	RNN	CNN	LSTM	STL BiLSTM-Att
Depression	Accuracy	93.20	88.10	89.10	91.00	91.80
	Macro-F1	93.20	88.10	89.10	91.00	91.80
Mental Health	Accuracy	81.00	68.10	81.00	83.30	84.00
	Macro-F1	78.20	63.10	78.30	80.80	81.60
Suicide	Accuracy	92.80	86.30	89.70	91.60	90.90
	Macro-F1	92.80	86.30	89.70	91.60	90.90

V. Discussion

- Multi-task learning lets the three tasks share one encoder, which acts as a regularizer and uses fewer parameters than three separate models.
- Pretrained GloVe embeddings provide a semantic initialization learned from large corpora, which helps with the informal language of online posts.

- The attention mechanism aids interpretability by focusing on relevant mental health-related words.
- Reporting results on a held-out test set, with duplicates removed before splitting, gives a more reliable estimate of generalization than evaluation on training data.
- Limitations: the three tasks come from different datasets rather than the same posts, labels are not clinically verified, and GloVe embeddings are static.
- Future work includes contextual embeddings (BERT, FastText), behavioral patterns and temporal analysis.

VI. Conclusion

This study presents an MTL attention-based hybrid model with pretrained embeddings for mental health prediction. A shared BiLSTM-attention encoder with three task-specific heads was trained and evaluated under a leakage-free protocol and compared with single-task baselines. The proposed framework lays the foundation for AI-driven mental health monitoring systems, enabling early detection and intervention. Future work includes expanding datasets, adding contextual embeddings such as BERT, integrating multi-modal analysis (text + voice), and real-world deployment.

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